

Regulatory Systems

Dr. Hader I. Sakr, Associate Professor, Medical Physiology | Source: 04_Regulatory_systems.pdf (28 slides) · Refs: Guyton & Hall 13th ed. (Unit II ch. 6; Unit V ch. 25); Ganong 25th ed. (Section I ch. 2, 5)

Legend: ★ = high-yield · [added] = not on slides ·  = the lecturer's handwritten annotation. Images marked "Slide N" come from the lecture; "Added diagram" = drawn for these notes.

 **Bridge from Lectures 01-03:** Lecture 01 defined homeostasis and its **SRACEER** loop. Lecture 03 listed intercellular communication (gap junction, neuronal, endocrine) without teaching it. This lecture names the **two systems that run homeostasis** (nervous and endocrine) and covers the communication methods. SRACEER returns here as the **reflex arc**.

1. Title / Objectives

Objective	Likely tested as
★ Compare the two regulatory systems	Nervous vs endocrine table (unit, onset, duration, example)
★ Structure and functional unit of the nervous system	Neuron parts, neuron types, myelin, reflex arc
★ List endocrine glands and their hormones	Pituitary-controlled vs not; mixed glands; part-time endocrine organs

Flow: Comparison table → nervous system (cells, neuron parts, axon, myelin, reflex arc) → endocrine system (chemical messengers, glands, hormones, other organs) → intercellular communication → conclusion → case.

2. Main Content (Lecture Order)

2.1 The Two Regulatory Systems (Slides 4-5, repeated 13 & 17)

★★★★ **Comparison (shown three times in the lecture, so it's the key slide):**

	Endocrine system	Nervous system
★ Structural unit	Ductless (endocrine) glands	Neurons and neuroglia
★ Functional unit	Chemical messengers (hormones)	★ Reflex arc
★ Onset	Rapid, moderate, or slow	★ Rapid
★ Duration	Short or prolonged	★ Short
Example	Metabolic functions	Muscular contractions

	<u>The endocrine system</u>	<u>The nervous system</u>
Structural unit	Ductless (endocrine glands)	Neurons and neuroglia
Functional unit	chemical messengers (hormones)	Reflex arc
Onset	Rapid, moderate or slow	Rapid
Duration	Short or Prolonged	Short
Example	Metabolic functions	Muscular contractions

Slide 17: Endocrine vs nervous system

 **Recap: Two systems.** Nervous = neurons/glia, reflex arc, rapid onset, short duration (e.g., muscle contraction). Endocrine = ductless glands, hormones, variable onset and duration (e.g., metabolism).

 **Link:** The lecture takes the fast system first.

 **Up next: The nervous system.**

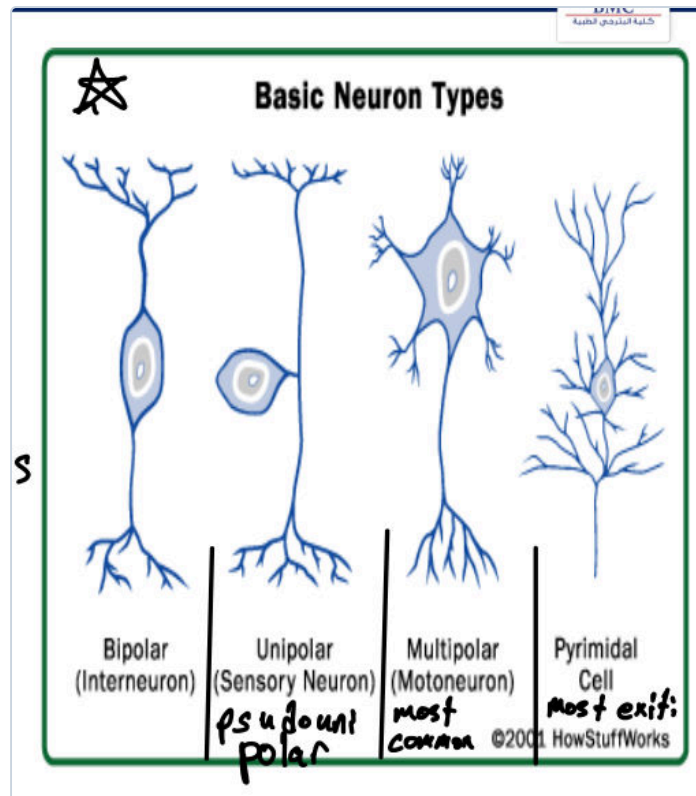
2.2 Nervous System: Cells & the Neuron (Slides 6-9)

Two cell types (billions of cells):

1. ★ **Neurons:** the **basic structural unit** of nervous tissue, specialised to **transmit action potentials**
2. ★ **Glial cells (neuroglia): supporting cells** (Greek "glue")  The lecturer wrote that these together are "**the structure unit of NS**".

★ Neuron types (Slide 7 figure, starred):

Type	Example (figure)	 Lecturer's note
Bipolar	Interneuron	
Unipolar	Sensory neuron	 Pseudounipolar
★ Multipolar	★ Motor neuron (typical spinal motor neuron)	 Most common
Pyramidal	Cortical cell	 partly illegible ("most exi...", possibly "most excitable")



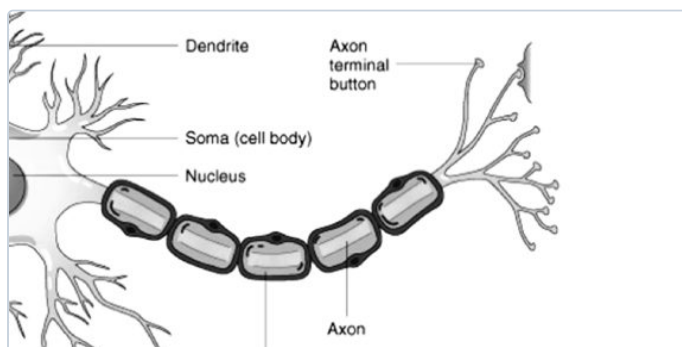
Slide 7: Basic neuron types with the lecturer's notes

★★ Parts of a typical (multipolar) neuron:

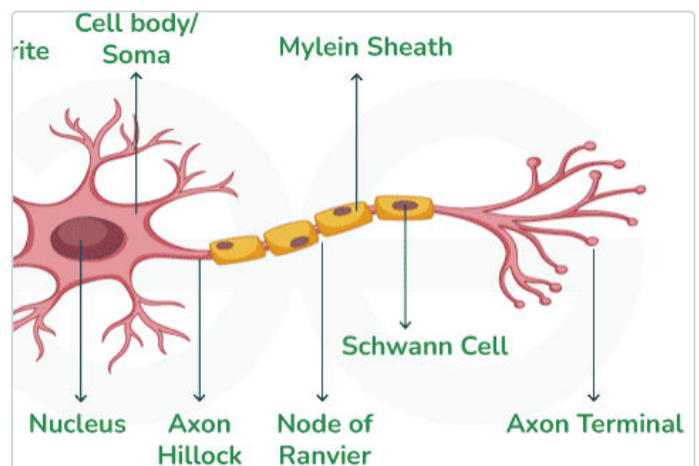
Part	Description
a. Soma (cell body)	Main body; ★ processing centre
b. Dendrites	Antenna-like processes; ↑ surface area for receiving signals
c. Axon (nerve fibre)	Single, elongated process

★ Axon details (Slide 9):

1. Arises from a thickened area of the soma: the ★ **axon hillock**
2. First portion = the ★ **initial segment** *[added: where action potentials usually start]*
3. **Axis cylinder** surrounded by plasma membrane (continuous with the cell membrane)
4. Ends in **synaptic knobs** (= **terminal buttons**), containing **vesicles of synaptic transmitter**
5. ★ Impulses travel **away from the cell body** along the axon



Slide 8: Neuron: dendrites, soma, axon, terminal buttons



Slide 9: Axon hillock, myelin sheath, nodes of Ranvier, Schwann cell

🔗 **Recap: Neuron.** Neurons (transmit) + glia (support) = the structural units. Multipolar (motor) is the most common type; the sensory neuron is pseudounipolar. Soma processes, dendrites receive, axon sends from hillock → initial segment → terminal buttons.

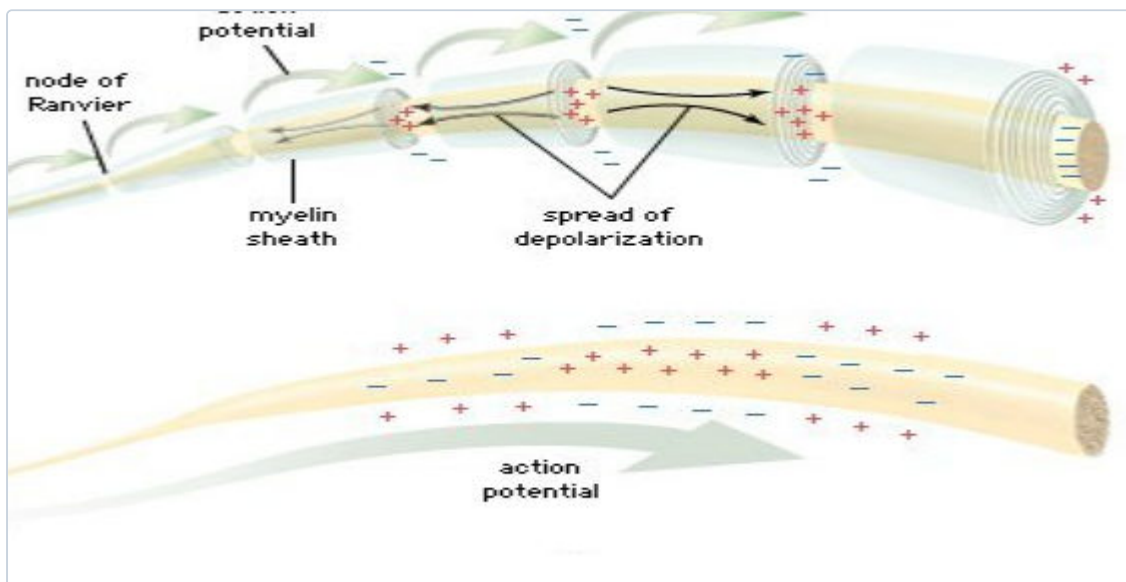
🔗 **Link:** How fast the axon conducts depends on its insulation.

▶ **Up next: Myelinated vs non-myelinated fibres.**

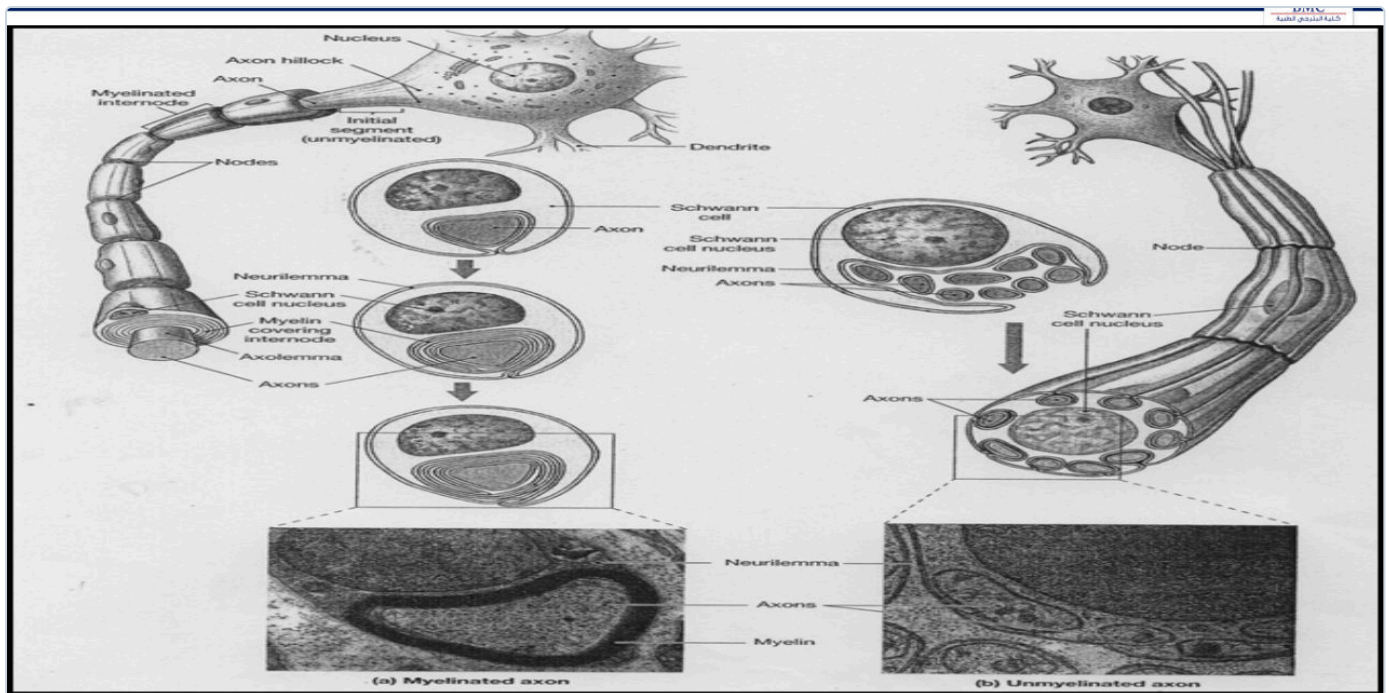
2.3 Myelinated vs Non-Myelinated Fibres (Slides 10-11)

	★ Myelinated	Non-myelinated
Sheath	★ Sphingomyelin myelin sheath: excellent insulator	Axon simply surrounded by Schwann cells, no myelin
Ion flow	★ ↓ ~5000-fold	Not reduced
Gaps	Covers the axon except at the nodes of Ranvier	n/a
Conduction	Fast (saltatory <i>[added]</i>)	Slow <i>[added]</i>

★ **Loss of myelin → delayed or blocked conduction.** *[added: e.g., multiple sclerosis, Guillain-Barré.]*



Slide 10: Myelinated (saltatory) vs non-myelinated conduction



Slide 11: Myelination by Schwann cells; myelinated vs unmyelinated fibres (EM)

🔁 **Recap: Myelin.** Sphingomyelin insulation cuts ion flow $\sim 5000\times$ except at nodes of Ranvier. Non-myelinated fibres sit in Schwann cells without myelin. Demyelination slows or blocks conduction.

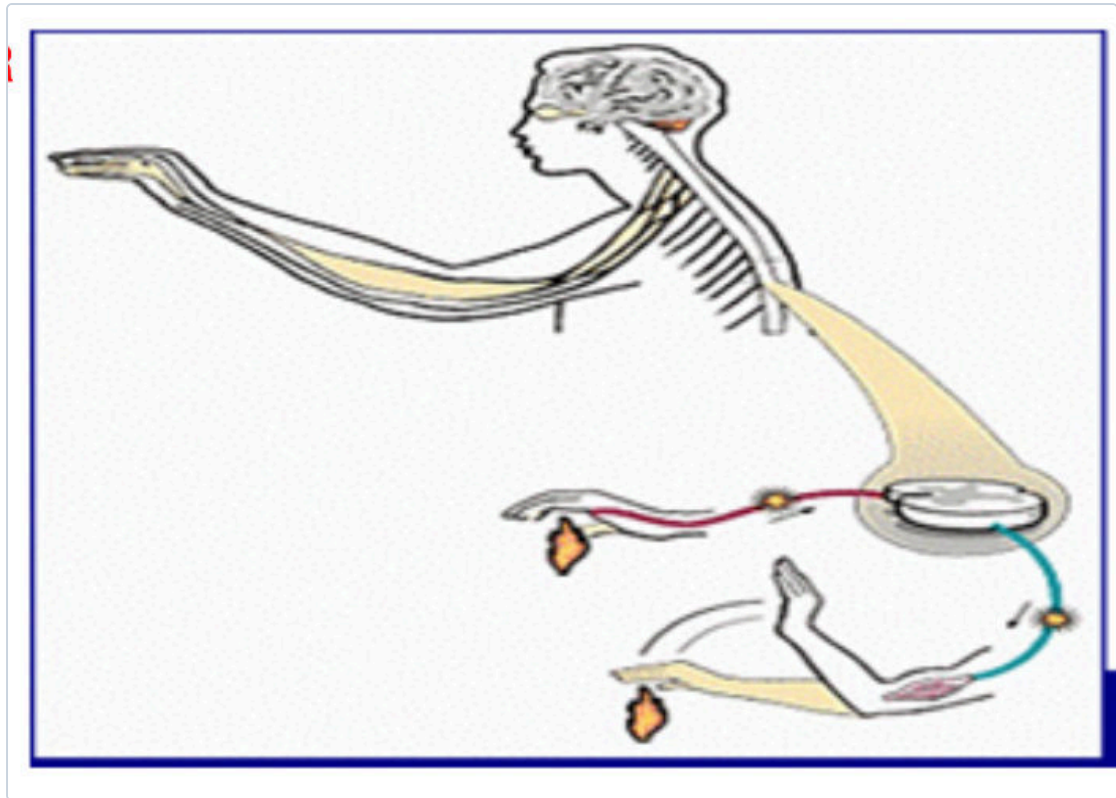
🔗 **Link:** Neurons are the structural unit; the functional unit is the circuit they form.

👉 **Up next: The reflex arc.**

2.4 Functional Unit: The Reflex Arc (Slide 12)

★★ **Reflex arc = the functional unit of the nervous system** (📝 lecturer wrote "**homeostasis**" beside it). Components = **SRACEER** (same mnemonic as the homeostasis loop in Lecture 01):

Letter	Component
S	A proper Stimulus
R	Sense organ (Receptor)
A	Afferent neuron
C	One or more synapses (Centre)
E	Efferent neuron
E	Effector organ
R	Proper Response



Slide 12: Reflex arc: withdrawal from a painful stimulus

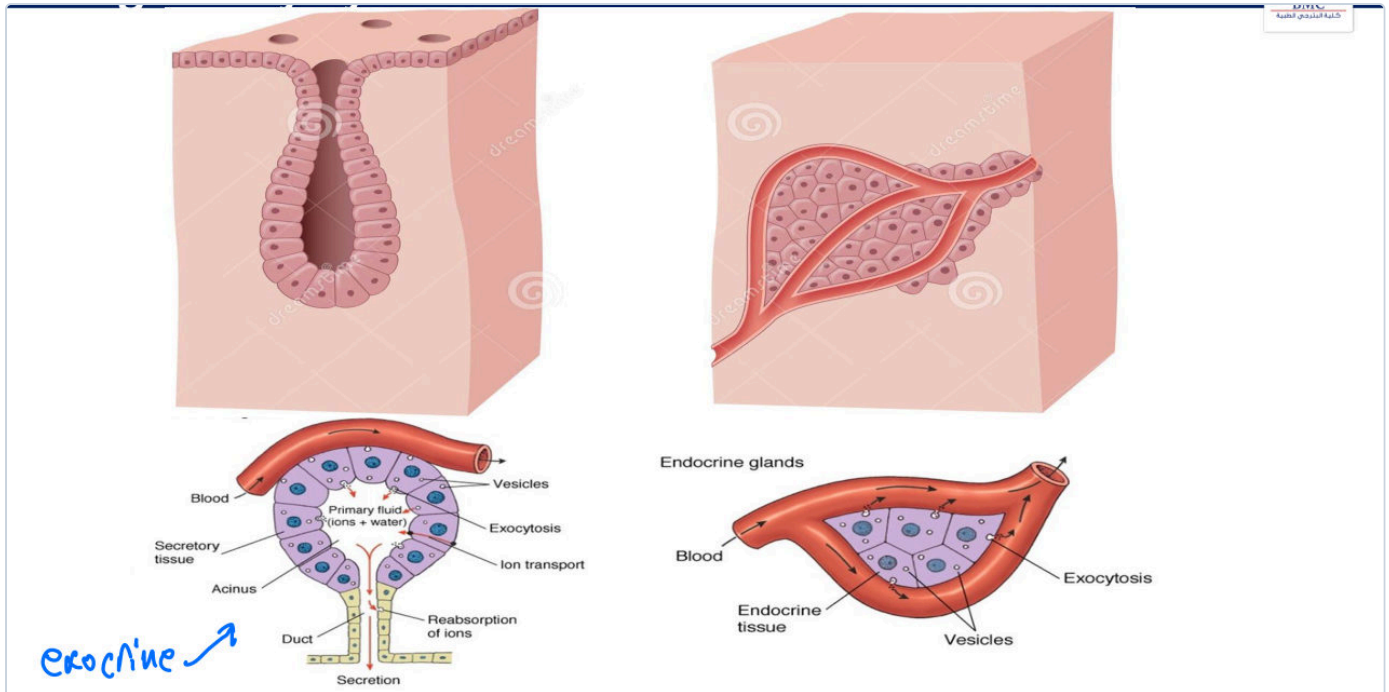
🔄 **Recap: Reflex arc.** SRACEER, the homeostasis loop built from neurons. The centre is one or more synapses.

🔗 **Link:** The second, slower regulatory system uses chemical messengers instead of wires.

▶ **Up next: The endocrine system.**

2.5 Endocrine System & Chemical Messengers (Slides 14-16)

★ **Endocrine** glands are **ductless**: they secrete into the blood. [added: *exocrine glands secrete through ducts.*] Slide 15 compares the two (📝 "exocrine" labelled on the figure).



Slide 15: Exocrine (with duct) vs endocrine (ductless) glands

- Endocrine glands are a ★ **slow control system**
- They act by **secreting hormones**

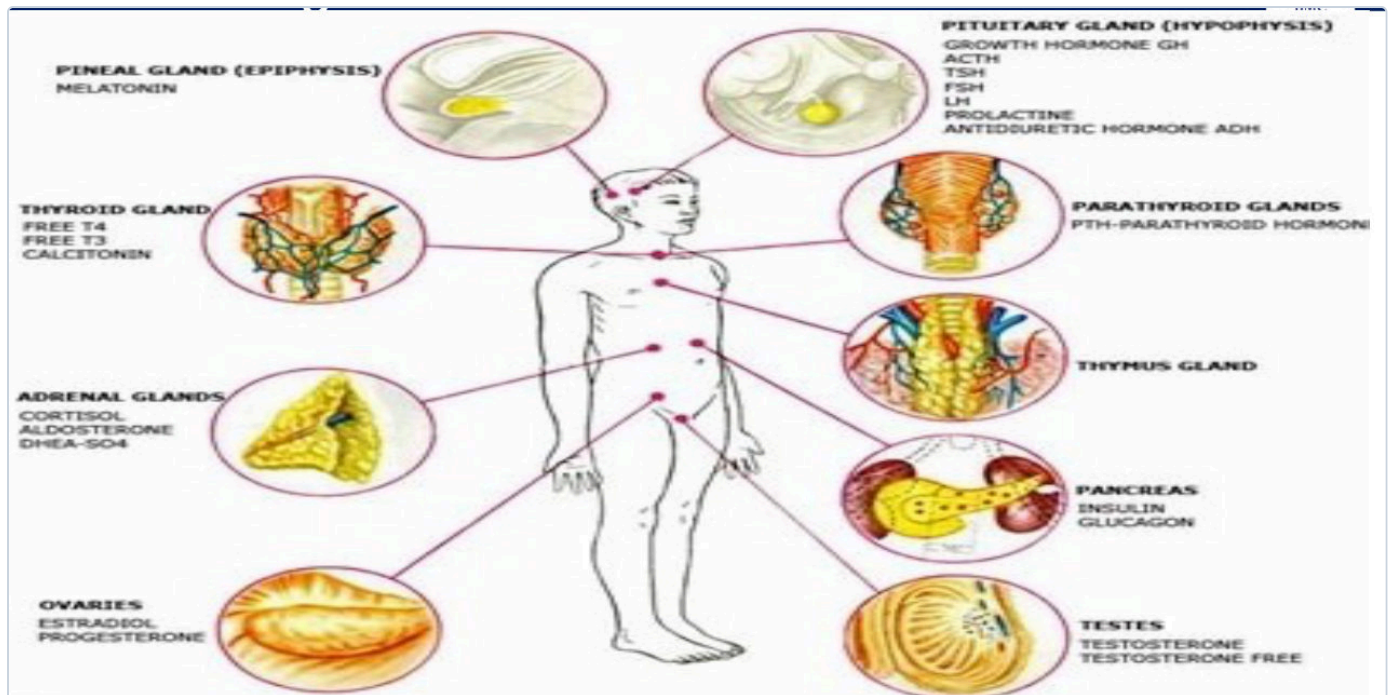
★★ Coordination by chemical messengers (Slide 16):

#	Messenger	Speed / scope (slide + 📝)
1	Gap junctions	📝 Direct, fast
2	Neurotransmitters (synaptic)	★ Rapid mechanism
3	Endocrine hormones	📝 Slow, systemic
4	Paracrine	📝 Through ISF (acts on neighbouring cells)
5	Autocrine	Acts on the same cell that secreted it [definition added]
6	Neuroendocrine hormones	★ Intermediate mechanism

2.6 Endocrine Glands & Their Hormones (Slides 18-20)

★★ Glands and hormones (Slide 18 figure):

Gland	Hormones (figure)
Pituitary (hypophysis)	GH, ACTH, TSH, FSH, LH, prolactin, ADH
Pineal (epiphysis)	Melatonin
Thyroid	Free T4, free T3, calcitonin
Parathyroids	PTH
Thymus	(named only)
Adrenals	Cortisol, aldosterone, DHEA-SO4
Pancreas	Insulin, glucagon
Ovaries	Estradiol, progesterone
Testes	Testosterone (total and free)



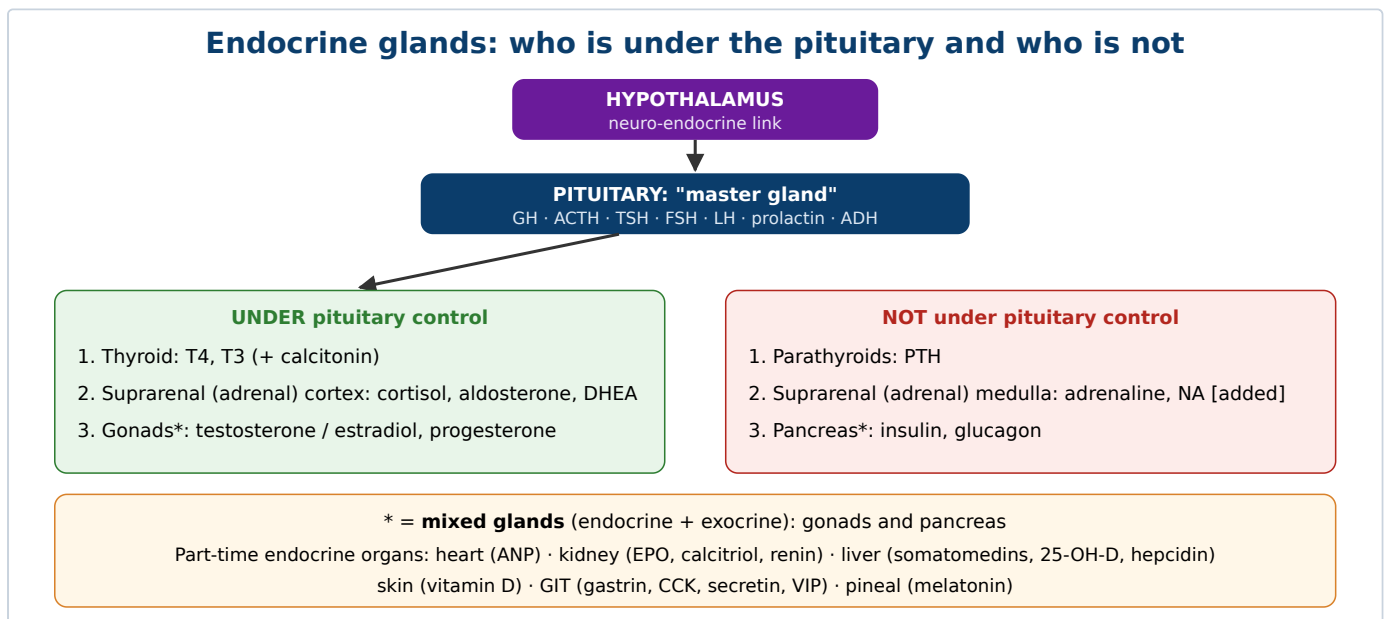
Slide 18: Endocrine glands of the body and their hormones

★★★★ Control by the pituitary (Slide 19, starred):

- ★ **Pituitary = the master gland**
- ★ Pituitary is controlled by the **hypothalamus (neuro-endocrine)**

Under pituitary control	NOT under pituitary control
1. Thyroid	1. Parathyroids
2. Suprarenal (adrenal) cortex	2. Suprarenal (adrenal) medulla
3. Gonads (testes, ovaries) †	3. Pancreas †

✎ † = **mixed glands** (endocrine + exocrine): **gonads** and **pancreas**.



Added diagram: which glands the pituitary controls

★★★ Organs with endocrine function alongside their main job (Slide 20):

Organ	Hormone(s)
★ Heart	Atrial natriuretic peptide (ANP)
★ Kidney	Erythropoietin, 1,25(OH)₂ cholecalciferol, renin , among others
★ Liver	Somatomedins, 25-hydroxycholecalciferol, hepcidin
Skin	Calciferol (vitamin D)
GIT	Gastrin, CCK, secretin, VIP , etc.
Pineal	Melatonin

Pattern [added]: vitamin D is made in **skin** → **25-hydroxylated in liver** → **1-hydroxylated in kidney** (active 1,25(OH)₂D). All three steps appear on this slide.

🔁 **Recap: Endocrine.** Ductless, slow. Messengers range from fast (gap junction, synaptic) through intermediate (neuroendocrine) to slow systemic (endocrine); paracrine and autocrine act locally. Hypothalamus → pituitary → thyroid, adrenal cortex, gonads. Parathyroid, adrenal medulla and pancreas are independent. Gonads and pancreas are mixed glands. Heart, kidney, liver, skin, GIT and pineal also secrete hormones.

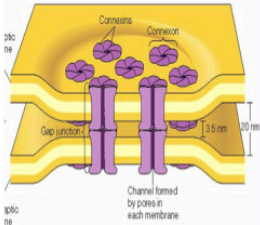
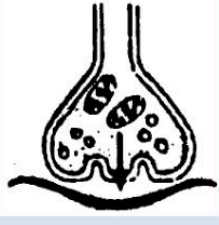
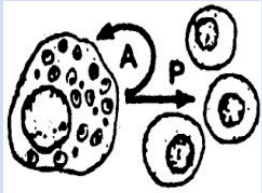
🔗 **Link:** The lecture ends by comparing all the ways cells talk to each other.

▶ **Up next: Intercellular communication.**

2.7 Intercellular Communication (Slides 21-22)

★★★★

	Gap junctions	Synaptic	Paracrine & autocrine	Endocrine
Route	Directly cell to cell	Across the synaptic cleft	By diffusion in ISF	By circulating body fluids
Range	Local	Local	Local	★ Systemic (📌 starred)

Gap Junctions	Synaptic	Paracrine & Autocrine	Endocrine
			
Directly from cell to cell	Across synaptic cleft	By diffusion in ISF	By circulating body fluids
Local	Local	Local	Systemic ★

Slide 22: Four types of intercellular communication

 **Recap: Communication.** Gap junction (direct), synaptic (cleft), paracrine/autocrine (ISF) are all local. Endocrine (blood) is the only systemic one.

 **Link:** Conclusion + case.

2.8 Conclusion (Slide 24)

- The **endocrine and nervous** systems are the **two regulatory systems**
- Nervous system: **structural units = neurons + neuroglia, functional unit = reflex arc**
- Different methods of chemical transmission between cells exist
- The body has several endocrine glands
- ★ **The endocrine and nervous systems are interconnected** (e.g., hypothalamus → pituitary)

3. High-Yield Comparison Tables

Table A: Nervous vs endocrine (see §2.1)

Table B: Neuron types

Type	Processes	Example
Bipolar	1 dendrite + 1 axon	Interneuron (per slide) <i>[added: retina, olfactory]</i>
Unipolar / pseudounipolar	Single process that splits	Sensory neuron (dorsal root ganglion <i>[added]</i>)
Multipolar	Many dendrites + 1 axon	Motor neuron, most common
Pyramidal	Pyramid-shaped soma	Cerebral cortex <i>[added]</i>

Table C: Chemical messengers by speed

Fast	Intermediate	Slow
Gap junctions, neurotransmitters	Neuroendocrine hormones	Endocrine hormones

Table D: Pituitary control (see §2.6)

Table E: Communication types (see §2.7)

4. Rules, Exceptions, Patterns

- **Nervous = rapid onset, short duration. Endocrine = variable onset, short or prolonged duration.**
- **Structural unit of NS = neurons + glia; functional unit = reflex arc.**
- **Impulses leave the soma via the axon** (hillock → initial segment → terminal buttons).
- **Myelin (sphingomyelin) cuts ion flow ~5000×** except at nodes of Ranvier.
- **Reflex arc = SRACEER**, the same letters as the homeostasis loop.
- **Pituitary controls thyroid, adrenal cortex, gonads. Not parathyroids, adrenal medulla, pancreas.**
- **Cortex yes, medulla no:** the adrenal is split between the two lists.
- **Mixed glands: gonads + pancreas.**

- **Only endocrine communication is systemic.**

5. Clinical Correlations *[added; the lecture has none]*

- **Multiple sclerosis** (CNS) and **Guillain-Barré** (PNS) = demyelination → slowed/blocked conduction.
- **Pituitary tumour/failure** affects thyroid, adrenal cortex and gonads, but **not** PTH, adrenaline or insulin.
- **Chronic kidney disease** → ↓ erythropoietin (anaemia) and ↓ 1,25(OH)₂D (bone disease).
- **Heart failure** → ↑ ANP/BNP (natriuretic peptides, used as markers).
- **Withdrawal reflex** (lecture case) happens before the pain is consciously felt.

6. Rapid Revision (Last-Day)

- Two regulatory systems: **nervous** (rapid, short, muscle) and **endocrine** (rapid-to-slow, short-or-prolonged, metabolism).
- NS structural units: **neurons + neuroglia**; functional unit: **reflex arc**.
- Neuron types: bipolar, **unipolar (pseudounipolar, sensory)**, **multipolar (motor, most common)**, pyramidal.
- Neuron: **soma** (processing), **dendrites** (receive, ↑ area), **axon** (hillock → initial segment → terminal buttons with transmitter vesicles). Impulses go **away from** the soma.
- Myelin = **sphingomyelin**, ↓ ion flow **5000x**, gaps = **nodes of Ranvier**; non-myelinated = Schwann cells without myelin.
- Reflex arc **SRACEER**.
- Messengers: gap junction (direct, fast), synaptic (rapid), endocrine (slow, systemic), paracrine (ISF), autocrine, neuroendocrine (intermediate).
- Pituitary (master) ← hypothalamus. **Controls thyroid, adrenal cortex, gonads; not parathyroid, adrenal medulla, pancreas. Mixed: gonads, pancreas.**
- Pituitary hormones: **GH, ACTH, TSH, FSH, LH, prolactin, ADH**.
- Other organs: **heart ANP; kidney EPO, 1,25(OH)₂D, renin; liver somatomedins, 25-OH-D, hepcidin; skin vit D; GIT gastrin, CCK, secretin, VIP; pineal melatonin.**
- Communication: gap junction, synaptic, paracrine/autocrine = **local**; endocrine = **systemic**.

7. Exam-Style Questions

Q1 (MCQ, lecture case, Slide 25). A child touches a hot stove and withdraws her hand within a fraction of a second; the response is brief. Which system, and why? 1) Endocrine, hormones act rapidly and briefly 2) **Nervous, its onset and duration are both rapid and short** 3) Endocrine, it controls skeletal muscle 4) Both equally, with identical timing **Answer: 2** (highlighted on the slide). Reflex arc = rapid onset, short duration; muscle contraction is the nervous system's example.

Q2 (MCQ). Which gland is **not** under pituitary control? a) Thyroid b) Adrenal cortex c) Testis **d) Adrenal medulla** **Answer: d.** Also parathyroids and pancreas.

Q3 (MCQ). The functional unit of the nervous system is: a) Neuron b) Neuroglia **c) Reflex arc** d) Synapse **Answer: c.** Neurons and glia are the **structural** units.

Q4 (MCQ). Which communication method is **systemic**? a) Gap junction b) Synaptic c) Paracrine **d) Endocrine** **Answer: d.**

Q5 (Short answer). Name the hormone produced by the heart, three by the kidney and three by the liver, per the lecture. **Answer:** Heart: **ANP**. Kidney: **erythropoietin, 1,25(OH)₂ cholecalciferol, renin**. Liver: **somatomedins, 25-hydroxycholecalciferol, hepcidin**.

8. Exam Pitfalls & Traps

1. **Structural vs functional unit of NS:** structural = **neurons + glia**; functional = **reflex arc**.
2. **Endocrine is always slow.** The table says onset can be **rapid, moderate or slow**, and duration **short or prolonged**.
3. **Adrenal gland:** cortex **is** under pituitary control, medulla **is not**.
4. **Mixed glands** = gonads and pancreas (both endocrine + exocrine).
5. **Paracrine vs autocrine:** paracrine → neighbouring cells via ISF; autocrine → the secreting cell itself.
6. **Impulse direction:** along the axon, **away from** the soma.
7. **Myelin covers the nodes of Ranvier.** No, the nodes are the gaps.
8. **Neuroendocrine = intermediate speed**, between synaptic (rapid) and endocrine (slow).
9. ⚠ **Handwriting:** the note under "Pyramidal cell" is partly illegible; don't rely on it.
10. ⚠ **Slide text:** "Mylein" on Slide 9's figure = **myelin**; "acts asis" = **acts as**.